

Tutorial Sheet

B.Tech.(C.S.)

Session 2025-26

Subject: Theory of Automata and Formal Language

Course Code: BCSC0011

Ques. Based on Deterministic Finite Automaton:

1. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ which accepts all strings that starts with 'a'.
2. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ such that length of the string is at least 2.
3. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ such that length of the string is exactly 2.
4. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ such that length of the string is less than 2.
5. Construct a DFA over $\Sigma = \{a, b\}$ where $|w| \bmod 3 = 0$.
6. Construct a DFA over $\Sigma = \{a, b\}$ where $n_a(w) = 2$.
7. Construct a DFA over $\Sigma = \{a, b\}$ where $n_a(w) \bmod 2 = 0$.
8. Construct a DFA over alphabet $\Sigma = \{a, b\}$ which accepts all the strings containing even no. of 'a' and even no. of 'b'.
9. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ which when interpreted as binary number is divisible by 3.
10. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ which accepts all the strings containing at least two 0's and two 1's.
11. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ which accepts all the strings contains 'a'.
12. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ which accepts all the strings ends with 'b'.
13. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ which accepts all the strings that starts with 01.
14. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ which accepts all the strings containing 00.
15. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ which accepts all the strings that ends with 10.
16. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ which accepts string starts with 'a' and ends with 'b'.
17. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ which accepts all strings that starts and ends with different symbols.
18. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ which accepts all strings that starts and ends with same symbol.
19. Construct a DFA over alphabet $\Sigma = \{a, b\}$ where every a is followed by b.
20. Construct a DFA over alphabet $\Sigma = \{0, 1\}$ where every 0 is not followed by 1.
21. Construct a DFA over alphabet $\Sigma = \{0, 1\}$ where every 0 has to be followed by 11.
22. Construct a DFA over alphabet $\Sigma = \{a, b\}$ where every a must not be followed by bb.
23. Let $\Sigma = \{a, b\}$ and let $L = \{a^n b^m \mid n, m \geq 1\}$. Design a DFA for L.
24. Let $\Sigma = \{a, b, c\}$ and $L = \{a^n b^m c^l \mid n, m, l \geq 1\}$. Design a DFA for L.
25. Let $\Sigma = \{0, 1, 2\}$ and $L = \{0^n 1^m 2^l \mid n, m, l \geq 0\}$. Design a DFA for L.
26. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ such that the second symbol from the LHS is 'a'.
27. Construct a DFA for $L = \{a^n \mid n \geq 0, n \neq 3\}$ where $\Sigma = \{a\}$.

28. Construct a DFA over the alphabet $\Sigma = \{0, 1\}$ such that the second symbol from the RHS is '1'.
29. Design a DFA for the language $L = \{ w : n_a = 1, w \in \{a, b\}^* \}$.
30. Design a DFA for the language $L = \{ w : n_a(w) \geq 1, w \in \{a, b\}^* \}$.
31. Design a DFA for the language $L = \{ w : n_a(w) \leq 3, w \in \{a, b\}^* \}$.
32. Construct a DFA which accept all strings that contains exactly two 0's and two 1's where $\Sigma = \{0, 1\}$.
33. Construct a DFA which accepts all the strings which are divisible by 3 where input is decimal number.
34. Construct a DFA for the following language, specified by a transition diagram, where $\Sigma = \{0, 1\}$.
 $L = \{ w \mid w \text{ begins with 0 or ends with 0 but not both} \}$.
35. Explain all the tuples of DFA.
36. What are the applications of Finite Automata.

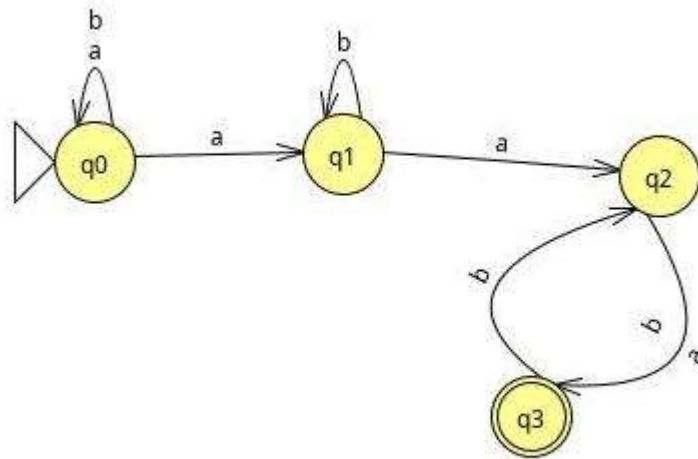
Ques. Based on Non-Deterministic Finite Automaton:

1. Design an NFA over the alphabet $\Sigma = \{a, b\}$ which accepts all strings start with a.
2. Design an NFA over $\Sigma = \{a, b\}$ which contains a.
3. Design an NFA over $\Sigma = \{a, b\}$ which ends with a.
4. Design an NFA over $\Sigma = \{a, b\}$ which starts with ab.
5. Design an NFA over $\Sigma = \{a, b\}$ which contains ab.
6. Design an NFA over $\Sigma = \{a, b\}$ which ends with ab.
7. Design an NFA accepting all strings over $\{0,1\}$ which ends with 1 but does not contain the substring 00.
8. Design an NFA which accepts all strings of length at least three and third symbol from right is 'a' where $\Sigma = \{a, b\}$.
9. Design an NFA over the alphabet $\Sigma = \{a, b\}$ and all the strings start with "ab".
10. Design an NFA over the alphabet $\Sigma = \{a, b\}$ and all the strings ends with "ab".
11. Design an NFA over the alphabet $\Sigma = \{a, b\}$ and all the strings containing "ab".
12. Design a NFA which accepts all the strings over $\{0,1\}$ that have at least two consecutive 0's or 1's.
13. Design an NFA which accepts $L = \{ ab, ba \}$.
14. Design an NFA accepting all strings over $\{0,1\}$ which ends with 1 but does not contain the substring 00.
15. Design an NFA over $\Sigma = \{a, b\}$ which accepts language $L = \{ w : |w| \geq 3 \text{ and third symbol of } w \text{ from right is 'a'} \}$.

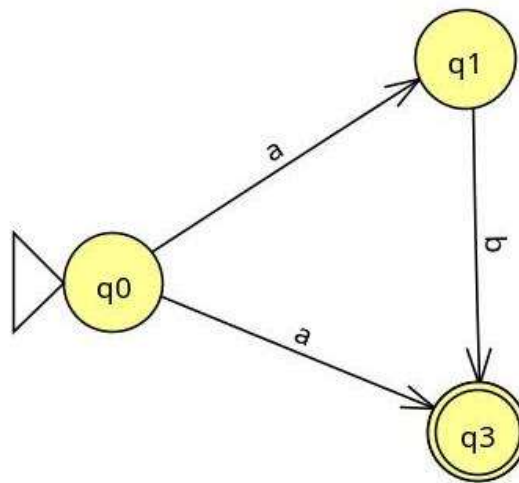
Ques. Based on NFA to DFA Conversion:

1. Design an NFA over the alphabet $\{0, 1\}$ that accepts all the strings starting with 00 and then convert it into an equivalent DFA.
2. Design an NFA over the alphabet $\{a, b\}$ that accepts all the strings containing aa or bb as substring and then convert it into an equivalent DFA.
3. Design an NFA over the alphabet $\{a, b\}$ that accepts all strings ending with ab and then convert it into an equivalent DFA.
4. Design an NFA over the alphabet $\{a, b\}$ which accepts the strings in which the second symbol from RHS is 'b'. Convert it into DFA.

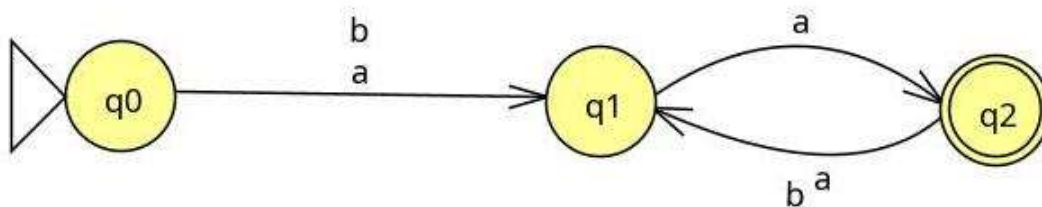
5. Design an NFA over the alphabet $\{0, 1\}$ that accepts all the strings in which the second symbol from LHS is '0' and last symbol is '1'. Convert it into DFA.
6. Convert the following NFA into DFA.



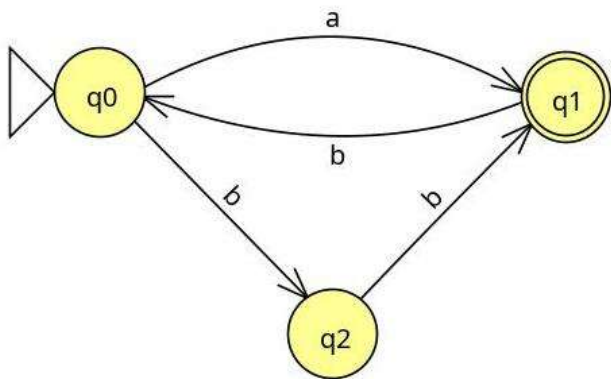
7. Convert the following NFA into DFA.



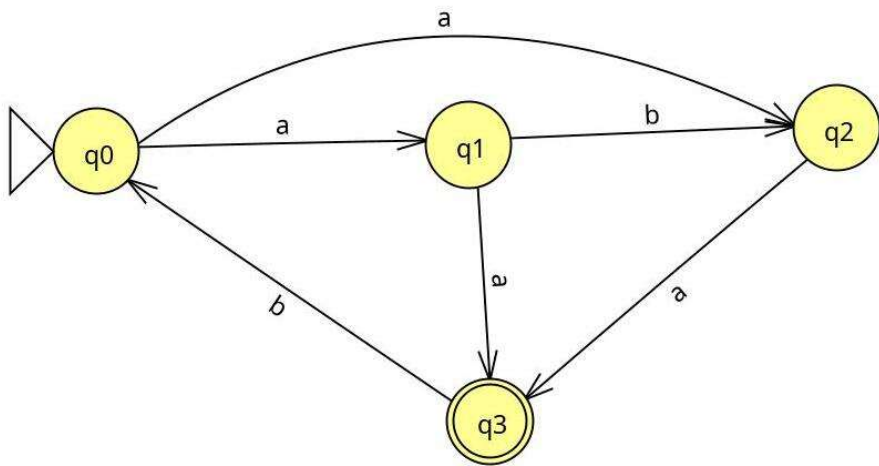
8. Convert the following NFA into DFA.



9. Convert the following NFA into DFA.

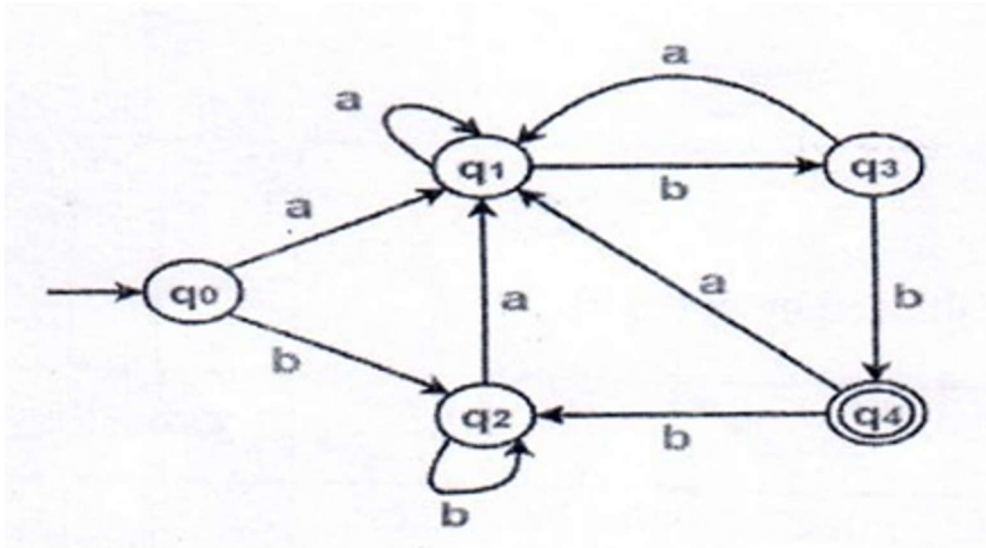


10. Convert the following NFA into DFA.

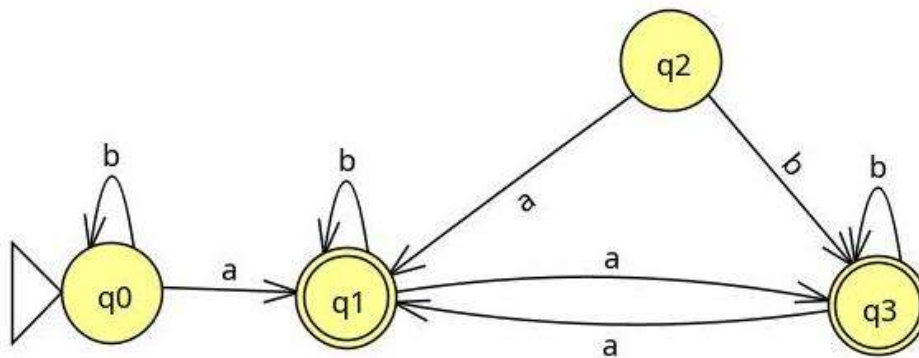


Ques. Based on DFA Minimization:

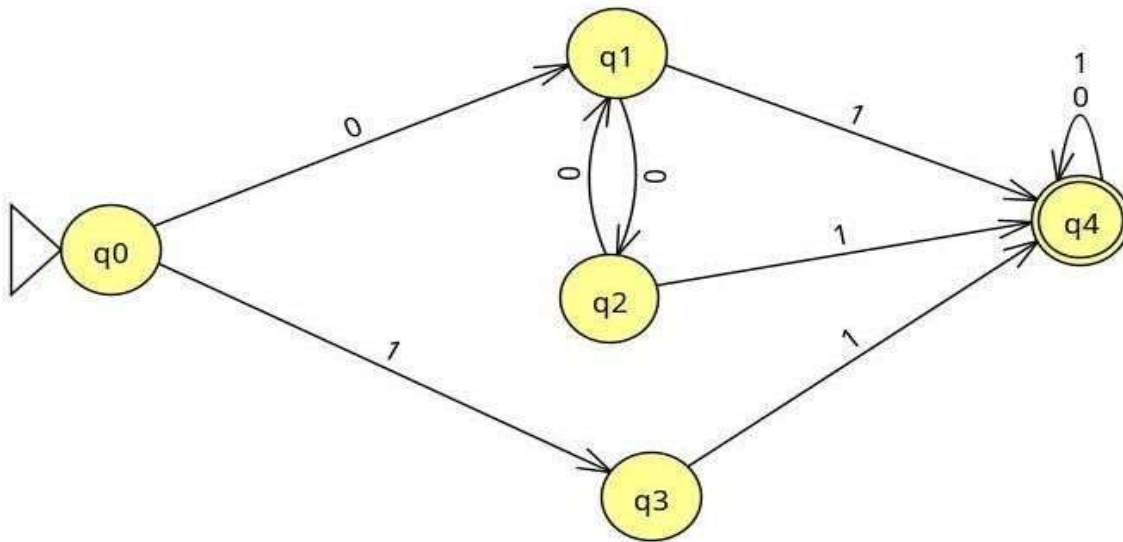
1. Minimize the following DFA.



2. Minimize the following given DFA.



3. Minimize the following given DFA.



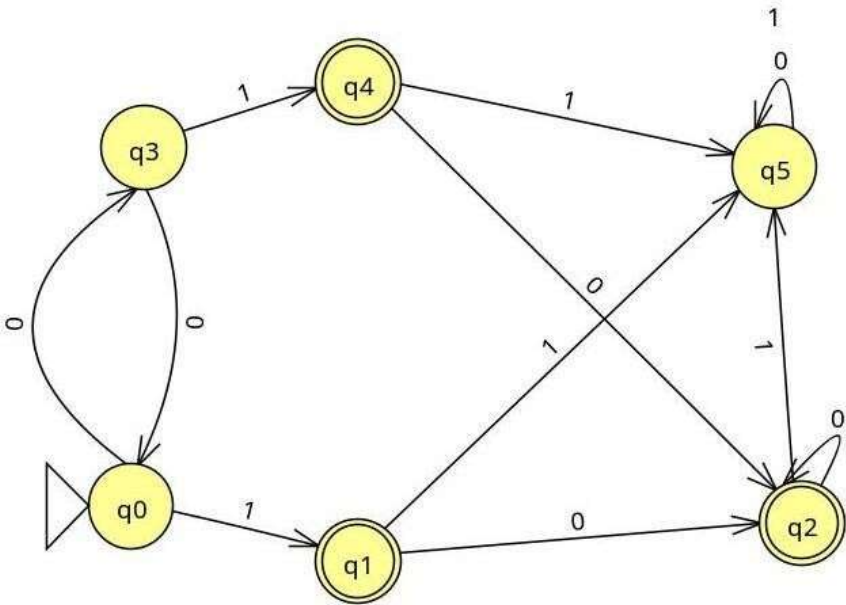
4. Minimize the DFA whose transition table is given below. Draw the minimized DFA as well.

Present State	Next State	
	a	b
→q0	q1	q5
q1	q6	q2
q2 (Final state)	q0	q2
q3	q2	q6
q4	q7	q5
q5	q2	q6
q6	q6	q4
q7	q6	q2

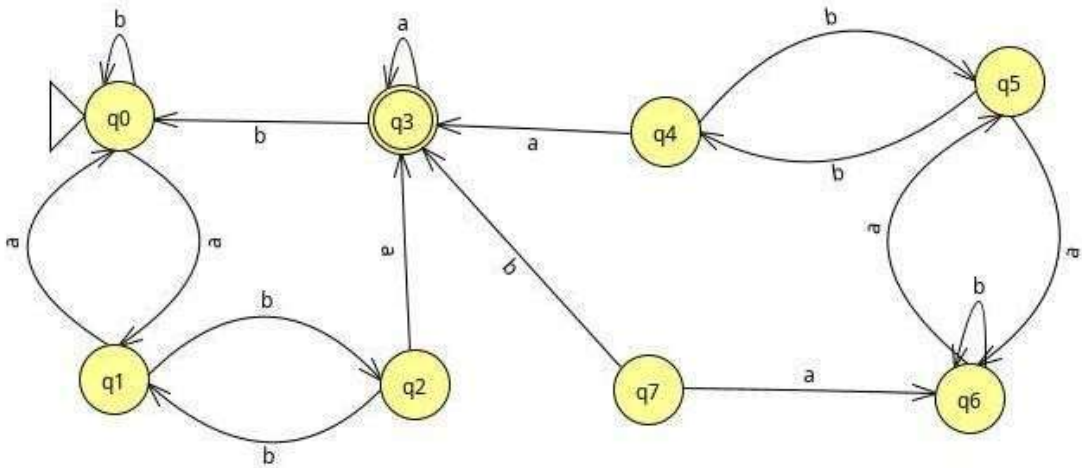
5. Minimize the DFA whose state transition table is given below:

States/input	0	1
→q ₀ +(final state)	q ₁	q ₃
q ₁	q ₄	q ₂
q ₂ +(final state)	q ₁	q ₅
q ₃	q ₀	q ₄
q ₄	q ₄	q ₄
q ₅	q ₂	q ₄

6. Minimize the following given DFA.

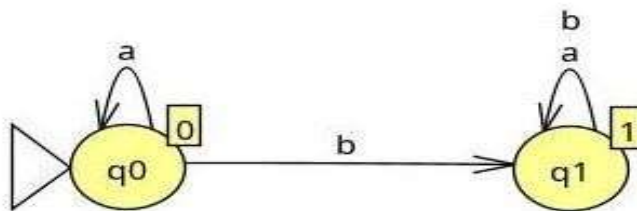


5. Minimize the following DFA.

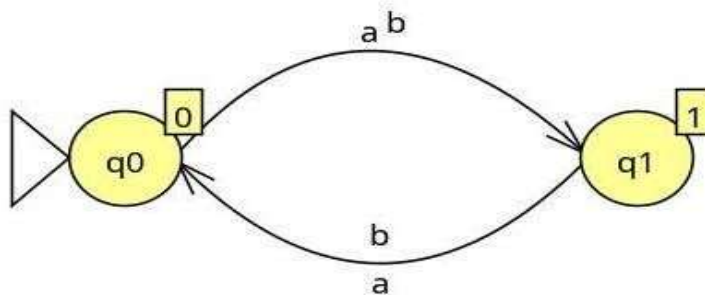


Ques. Based on Moore and Mealy Machine:

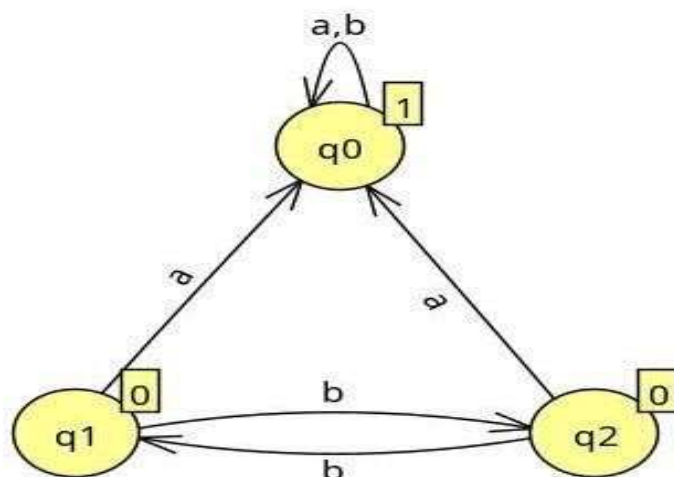
1. Design a Moore machine which counts the occurrence of substring "ab" in input string where $\Sigma = \{a, b\}$.
2. Design a Mealy machine, which prints 1's complement of input bit string over the alphabet $\Sigma = \{0,1\}$.
3. Design a Mealy machine, which prints 2's complement of input bit string over the alphabet $\Sigma = \{0,1\}$.
4. Design a Moore machine to compute 2's complement of a binary string.
5. Design a Moore machine which prints 1 for each occurrence of substring "baa" in input string where $\Sigma = \{a, b\}$.
6. Design a Moore machine that produces output 'X' for each occurrence of '01' in the input string, and produces an output 'Y' for each occurrence of '10' and produces output 'Z' for any other string.
7. Design a Moore machine which determine the residue mod 3 for each binary string treated as binary integer.
8. Convert the following Moore machine into Mealy Machine.



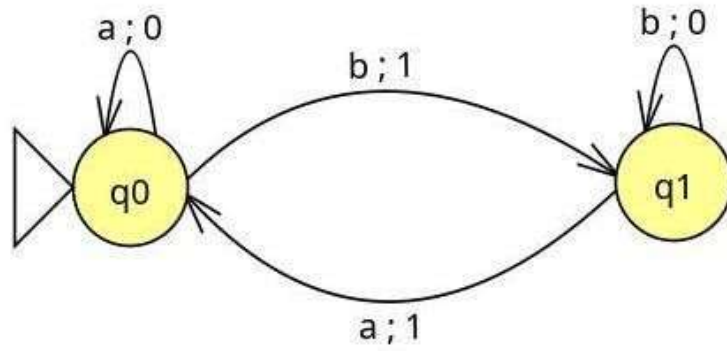
9. Convert the following Moore machine into Mealy Machine.



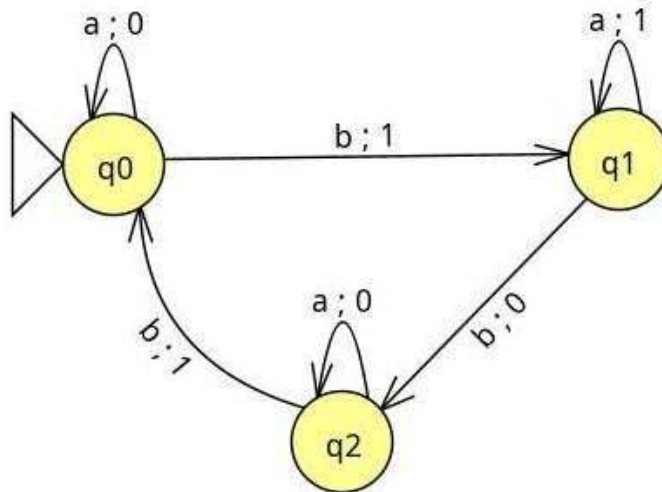
10. Convert the following Moore machine into Mealy Machine.



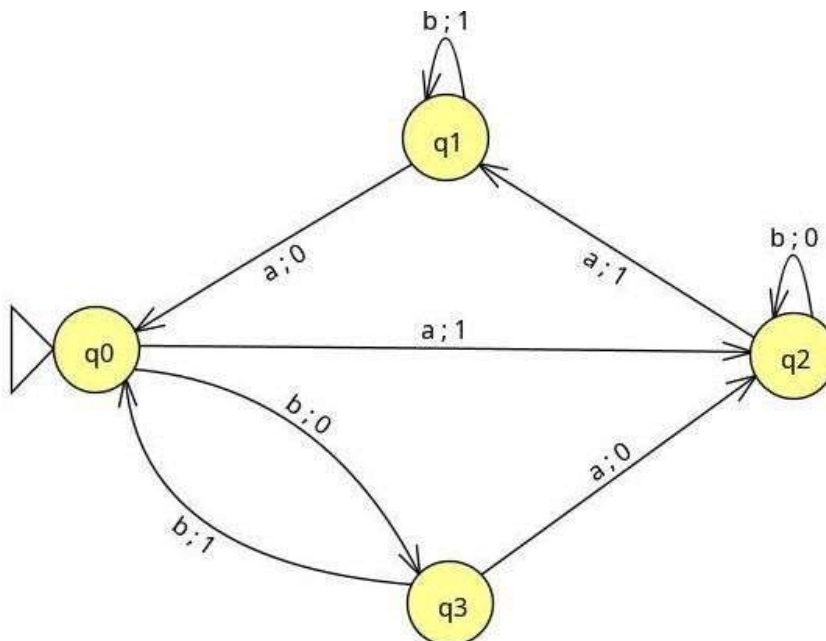
11. Convert the following Mealy machine into Moore machine.



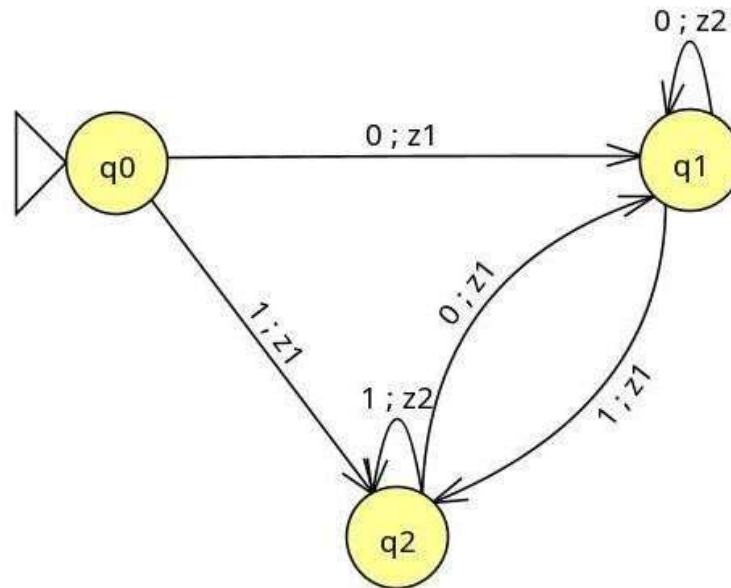
12. Convert the following Mealy machine into Moore machine.



13. Convert the following Mealy machine into Moore machine.



14. Convert the following Mealy machine into Moore machine.



Ques. Based on Regular Expression and Finite Automata:

- Write a regular expression for the below languages over alphabet $\Sigma = \{0, 1\}$:
 - All strings of length exactly 2.
 - All string of length at least 2.
 - All strings of length at most 2.
- Write the regular expression over alphabet $\{a, b, c\}$ containing at least one 'a' and at least one 'b'.
- Write the regular expression for the set of strings of 0's and 1's whose tenth symbol from the right end is '1'.
- Write the regular expression for the language:
 $L = \{ a b^n \mid n \geq 3, w \in \{a, b\}^* \}$.
- Write the regular expression for the language:
 $L = \{ w : |w| \bmod 3 = 0 \}, w \in \{a, b\}^* \}$.
- Find the regular expression for the following:
 - $L = \{ a^n b^m \mid n, m \geq 3 \}$
 - $L = \{ a^n \mid n \geq 0, n \neq 3 \}$.
- Write the regular expression for the following language over the alphabet $\{0, 1\}$:
 - All strings ending in "01".
 - All strings not ending in "01".
 - All strings containing an even number of 0's.
 - All strings not containing the substring "10".
- Write a regular expression for the language over the alphabet $\{0, 1\}$ consisting of all strings that contain at most two occurrences of the substring "00".
- Write a regular expression for the set of strings of 0's and 1's not containing 101 as a substring.

10. Write a regular expression for the set of strings containing at most three 0 over alphabet $\{0, 1\}$.
11. Write the regular expression for the set of strings which are ending with '1', but do not contain "00" where alphabet is 0 and 1.
12. Construct the finite automaton equivalent to the below regular expression.

$$a(a + b)^*$$
13. Construct the finite automata equivalent to the below regular expression.

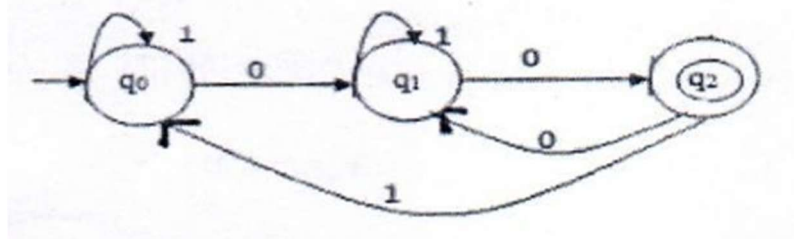
$$aa + ab + ba + bb$$
14. Construct the finite automaton equivalent to the below regular expression.

$$(0 + 1)^*(0 + 11)(0 + 1)^*$$
15. Construct the finite automaton equivalent to the below regular expression.

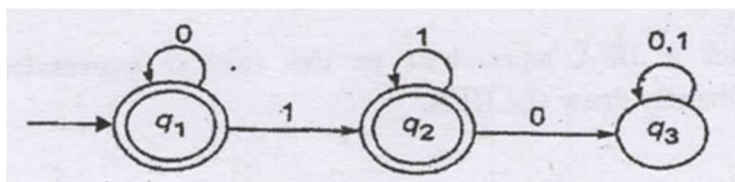
$$abb + a(a + b)^*ba$$
16. Construct the finite automaton equivalent to the below regular expression using null removal technique.

$$(a + b)(a + b)^*(a + b)$$
17. Construct the finite automata for the regular expression, RE = $(a + b)^*abb$.
18. Construct the finite automaton equivalent to the below regular expression.

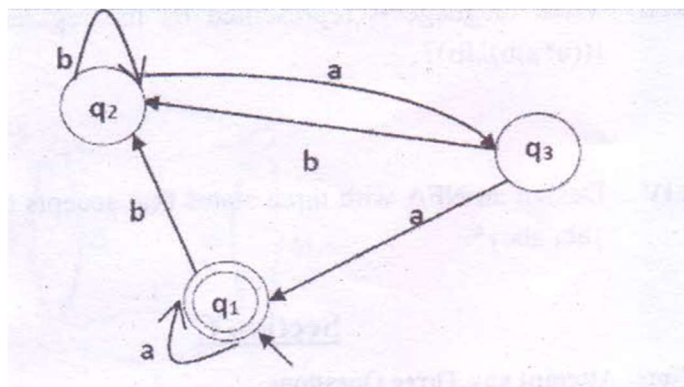
$$(a + b)^*(aa + bb)(a + b)^*$$
19. State and prove Arden's theorem.
20. Prove that:
 - a) $(r_1^*)^* = r_1^*$
 - b) $r_1^*(r_1 + r_2)^* = (r_1 + r_2)^*$
21. Find the Regular Expression corresponding to the following given FA using Arden's Theorem.



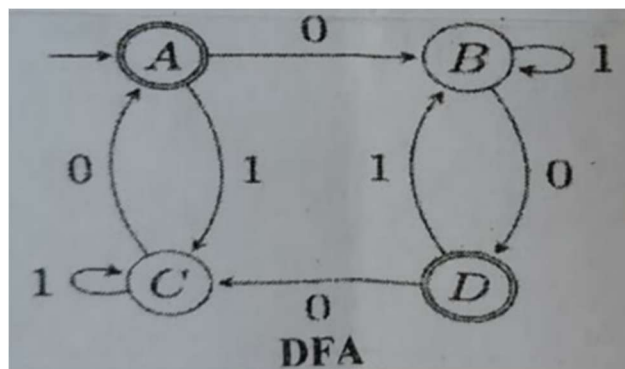
22. Find the Regular Expression equivalent to the below given DFA:



23. Find the Regular Expression equivalent to the below given DFA:



24. Find the Regular Expression equivalent to the below given DFA:



Ques. Based on Pumping Lemma theorem:

1. Prove that $L = \{ a^n b^n \mid n \geq 0 \}$ is not regular.
2. Show that $L = \{ a^p \mid p \text{ is a prime number} \}$ is not regular.
3. Show that $L = \{ ww \mid w \in \{a, b\}^* \}$ is not regular.
4. Show that $L = \{ a^n b^n c^n \mid n \geq 1 \}$ is not regular.
5. Show that $L = \{ w \in \{a, b\}^* \mid \#a(w) = \#b(w) \}$ is not regular.

Ques. Based on Pushdown Automata:

1. Design pushdown automata for language $L = \{ a^n b^n \mid n \geq 0 \}$.
2. Design pushdown automata for language $L = \{ a^n b^{2n} \mid n \geq 1 \}$.
3. Construct a PDA accepting Language $L = \{ wcw^R \mid w \in \{a, b\}^* \}$.
4. Construct a NPDA for $L = \{ ww^R \mid w \in \{a, b\}^* \}$.
5. Construct a PDA for $L = \{ a^n b^m c^n \mid m, n \geq 1 \}$.
6. Construct a PDA machine accepting $L = \{ a^i b^k c^k \mid i, k \geq 1 \}$.
7. Construct a PDA machine accepting $L = \{ a^i b^{j+i} c^j \mid i, j \geq 1 \}$.
8. Construct a PDA machine accepting $L = \{ a^m b^n c^{n+m} \mid n, m \geq 1 \}$.
9. Construct a PDA machine accepting $L = \{ a^i b^i c^j d^j \mid i, j \geq 1 \}$.
10. Construct a PDA machine accepting $L = \{ a^i b^j c^j d^i \mid i, j \geq 1 \}$.
11. Design a PDA for accepting the language $L = \{ a^m b^n c^p d^q \mid m+n=p+q, m, n, p, q \geq 1 \}$.